$$\Sigma_{\mathrm{SC}} = \Gamma_{\mathrm{NS}} \frac{\Delta(V_{\mathrm{Z}})\,\tau_x - \omega}{\sqrt{\Delta(V_{\mathrm{Z}})^2 - \omega^2}}, \qquad \Delta(V_{\mathrm{Z}}) = \Delta_0 \sqrt{1 - (V_{\mathrm{Z}}/V_C)^2}$$